

9/15 ch. 1.4

span  all linear combination of subspace span W.

ch. 1.5

Basis: Vectors in $\mathbb{R}^3$ form a basis of $\mathbb{R}^3$ and have two important properties.

- ① They span $\mathbb{R}^3$. They can be write an arbitrary vector as a linear combination.
- ② They are linearly independent which can't be decompose.

ex. $a[1,0,0] + b[0,1,0] + c[0,0,1] = [0,0,0]$

if only $a=0, b=0, c=0$ solution, the vectors linearly independent.

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Standard Basis of $\mathbb{R}^n$

Consider the vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ in $\mathbb{R}^3$. These vectors have two very important properties.

- (i) They *span* $\mathbb{R}^3$: That is, we can write an arbitrary vector (x, y, z) as a linear combination of the vectors as follows,

$$(x, y, z) = x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)$$

- (ii) They are *linearly independent*: That is, the identity

$$p(1, 0, 0) + q(0, 1, 0) + r(0, 0, 1) = (0, 0, 0)$$

for scalars p, q , and r , has the unique solution $p = 0, q = 0, r = 0$.

Dimension ($\mathbb{R}^n$) The standard Basis $\{[1, 0, \dots, 0], [0, 1, \dots, 0], \dots, [0, \dots, 1]\}$ of n vectors. The dimension of $\mathbb{R}^n$ is n

ch. 1.6

Dot product:  $\vec{u} = [a_1, a_2, a_3, \dots, a_n]$ $\vec{v} = [b_1, b_2, b_3, \dots, b_n]$ $\vec{u} \cdot \vec{v} = \underbrace{a_1b_1 + a_2b_2 + \dots + a_nb_n}_{\text{Real Number}}$

Properties of the Dot Product

Let $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$ and let c be a scalar. Then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
3. $c\mathbf{u} \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot c\mathbf{v}$
4. $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Norm: (length/Magnitude) $\vec{u} = [a_1, a_2]$ $|\vec{u}| = \sqrt{a_1^2 + a_2^2}$

Unit vector: vector whose norm is 1

Normalizing: let a vector become unit vector.

Angles between vector: (夾角) θ is an angle between $\vec{u}$ and $\vec{v} \Rightarrow \cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|}$

$$|\cos\theta| = \left| \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} \right| \leq 1$$